Premier League Analytics Project Report

Team: Schema Squad

Course: DATA 201

Academic Year: 2024-2025

Contents

1. Introduction

The **Premier League Football Analytics System** is a dual-database solution designed to manage and analyze football match data across multiple seasons of the English Premier League. This comprehensive system integrates a **transactional (OLTP) database** for match entry and event tracking, with an **analytical (OLAP) database** that powers trend analysis and performance evaluation across teams, referees, and seasons.

The **OLTP database** ensures integrity and reliability in match-day operations. It supports efficient ingestion and validation of structured football data such as teams, match events, player statistics, and betting odds. Each table follows a normalized design pattern to maintain referential integrity and ensure seamless updates.

In addition to core football data, the system incorporates comprehensive **betting-related information** through the Bookmakers, Markets, and BettingOdds tables. These components allow for deeper analysis of market dynamics, odds fluctuations, and bookmaker strategies. This fusion of match results with betting data enables the system to support predictive analytics, performance benchmarking, and odds evaluation.

The **OLAP database**, built on a star schema, transforms raw match data into strategic insights. With fact tables like MatchResult, TeamMatchStats, and LeagueSnapshot, it empowers stakeholders to explore seasonal performance trends, identify patterns in regional play, and evaluate referee influence over match outcomes.

This dual-database architecture, combined with an automated ETL pipeline, creates a scalable foundation for data-driven decision-making in sports analytics — benefiting football analysts, clubs, and fans alike.

2. Data Sources

The Premier League Analytics System is powered by a structured collection of historical match data, extracted from a reputable football statistics provider. The dataset spans ten consecutive seasons of the English Premier League, stored as CSV files and processed through a dedicated ETL pipeline to ensure consistency and referential integrity across all records.

Source

- Football-Data.co.uk — A publicly available repository of European football results and betting odds.

File List

Each file corresponds to a single Premier League season and adheres to a uniform schema:

- EPL_2019_2020.csv
- EPL_2020_2021.csv
- EPL_2021_2022.csv
- EPL_2022_2023.csv
- EPL_2023_2024.csv

Data Fields

These CSVs contain detailed information including:

- **Match metadata**: date, home/away teams, final scores
- **Referee and stadium information**
- **Team performance metrics**: goals, shots, cards, fouls, possession
- **Betting odds**: odds for home win, draw, away win from various bookmakers

ETL Relevance

The data from these CSVs is ingested through an ETL pipeline that performs:

- **Extraction**: CSV parsing and structural validation
- **Transformation**: metric derivation, type enforcement, foreign key mapping
- **Loading**: inserting cleaned records into the OLTP schema, then aggregating into OLAP fact tables

This structured ingestion process ensures high data quality for both operational tracking and analytical insights.

3. Application Design

The architecture of the Premier League Analytics System is built on a robust technology stack that ensures modularity, maintainability, and performance for both transactional and analytical use cases. The application design is layered, separating data ingestion, database interaction, GUI interfaces, and analytical reporting.

Tools & Technologies Used

- **Python**: Core language for ETL scripting, database operations, and GUI logic
- **MySQL**: Relational database system used for both OLTP and OLAP schemas

- **Jupyter Notebooks**: Environment for testing ETL pipelines and running analytical queries
- **PyQt5**: Used to design the **Graphical User Interface (GUI)** for data entry and user interactions
- **Pandas & NumPy**: Libraries for structured data processing
- **Matplotlib & Seaborn**: Optional visualization tools for reporting in notebooks

Packages & Dependencies

The following Python packages and dependencies were used across modules:

- pandas
- numpy
- mysql-connector-python
- PyQt5 and PyQt5-tools

These tools enabled:

- Reading and transforming large CSV files
- Executing parameterized SQL queries from scripts and GUI
- Designing custom dialog boxes and windows for user input (via PyQt)
- Visualizing trends and KPIs using Python plots

Module Overview

The application is organized into distinct modules:

- **ETL Module**
  Automates data ingestion and validation from raw CSVs into the operational database.
- **GUI Module**
  Built using PyQt5, it provides an interactive interface for users to:
  - Load and view match data
  - Add new records
  - Trigger ETL operations
  - Query team statistics and referee data
- **Operational Database (OLTP)**
  Normalized schema for storing raw football data and ensuring referential integrity.
- **Analytical Database (OLAP)**
  Star schema used for running complex queries and extracting meaningful insights.

- **Reporting Layer**

  Jupyter notebooks support in-depth analysis and data visualization.

- **Implementation Architecture & Workflow**

  The Premier League Analytics System employs a **hybrid approach** combining Jupyter notebooks for database schema setup with a Python GUI application for operational data management and analytics visualization.

  Key Implementation Points:

  - **Database Schema Creation**: Jupyter notebooks handle the initial OLTP and OLAP database setup
  - **Operational Data Management**: PyQt5 GUI handles day-to-day operations including CSV uploads and ETL processing
  - **Analytics Access**: Both GUI dashboards and Jupyter notebooks provide analytical capabilities
  - **User Authentication**: Secure login system (default: User id: Admin/ Password: admin) controls access to operational functions

4. Database Design

The Premier League Analytics System was designed using a **single MySQL database** that houses both the **operational (OLTP)** and **analytical (OLAP)** components. While logically separated in design, both schemas coexist within the same database environment to simplify data management and streamline the ETL process. The operational portion is structured using normalized tables to ensure referential integrity and minimize redundancy, while the analytical portion follows a star schema model to support efficient, high-performance querying and reporting.

4.1 Operational Database (OLTP)

The operational database section includes normalized tables that support the structured ingestion of raw match data, team statistics, referee assignments, and bookmaker odds across multiple seasons. This design ensures accurate, consistent data entry and prepares the foundation for downstream analytical processing.

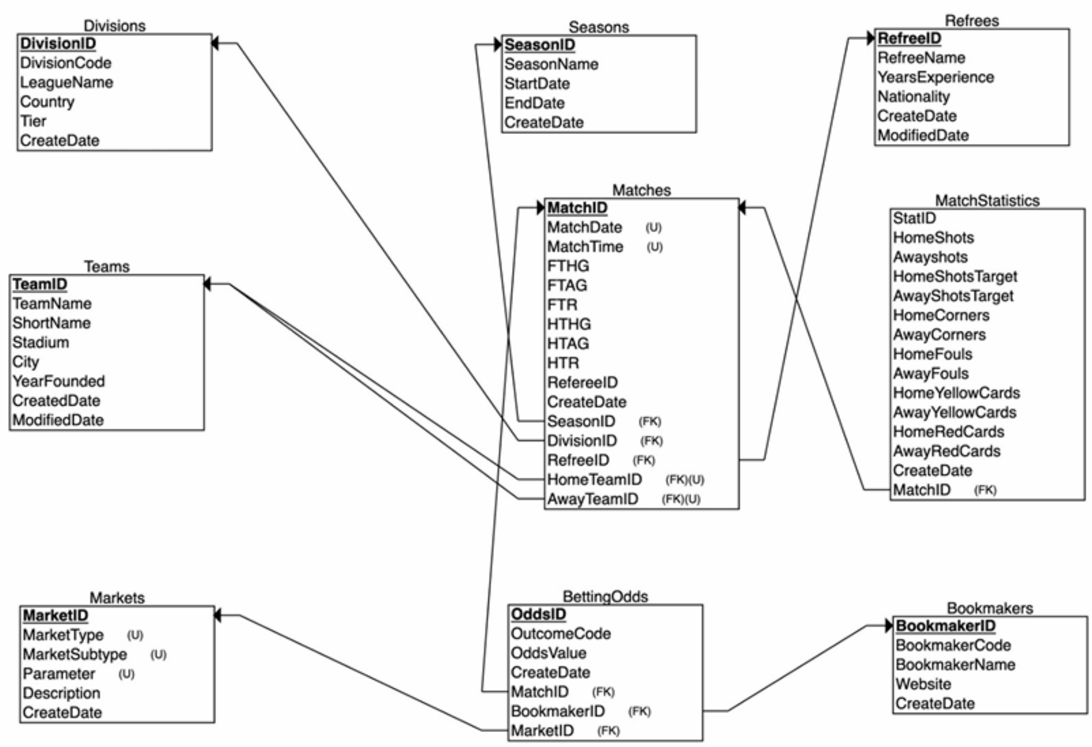
Key tables include:

- Teams
- Seasons
- Referees
- Divisions
- Matches

- MatchStatistics
- Bookmakers
- Markets
- BettingOdds

The structure is normalized up to at least **3NF (Third Normal Form)** and ensures:

- Referential integrity between matches and their metadata (teams, referees, seasons)
- Separation of static dimensions (e.g., team, season) from transactional data (e.g., matches)
- Accurate tracking of odds data from multiple bookmakers for each match



4.2 Analytical Database (OLAP)

The analytical database was built using a **star schema**, optimized for reporting and trend analysis. It includes multiple fact tables and dimension tables to enable slicing and dicing of data across time, location, referee, and team performance.

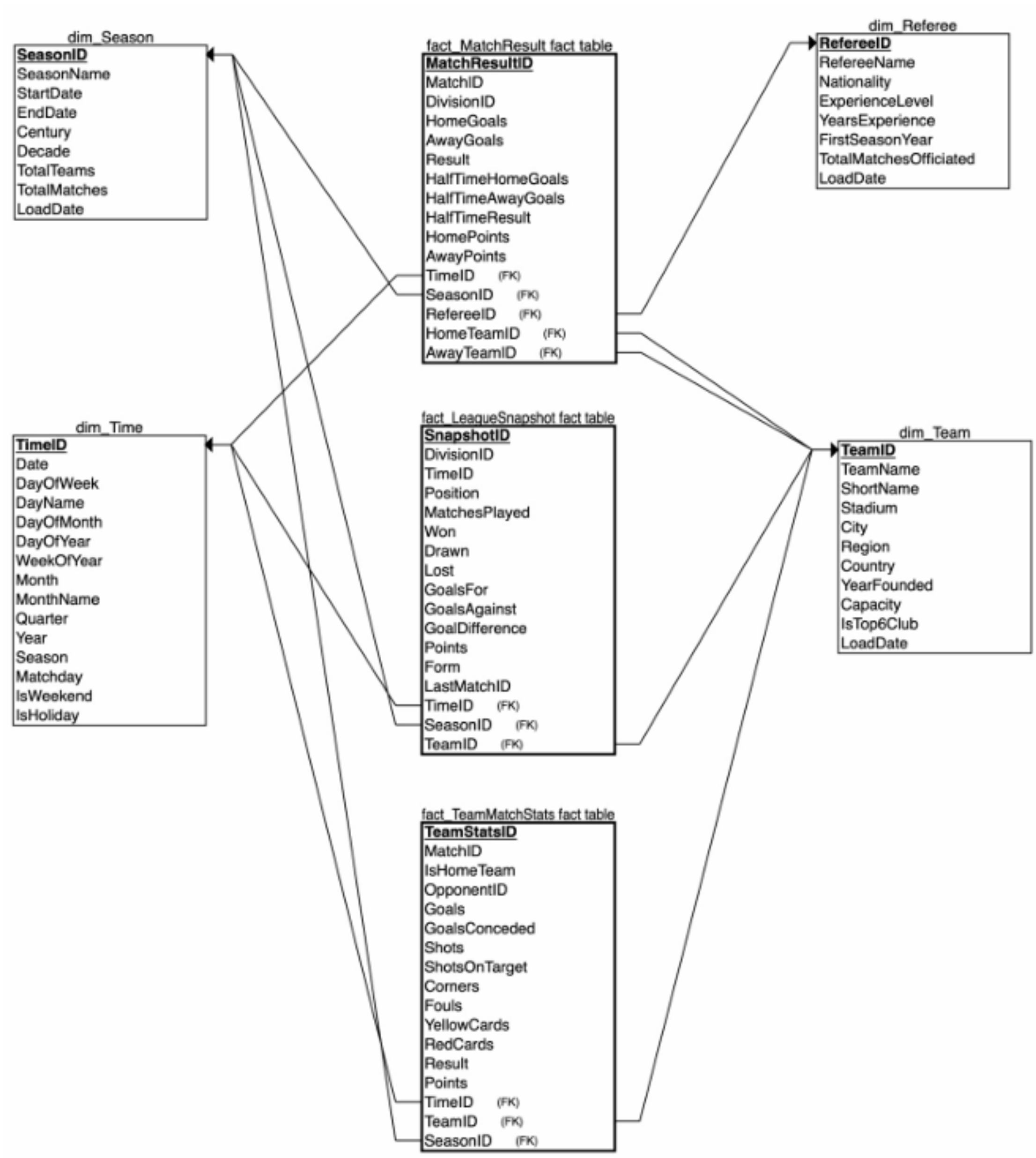
Fact Tables:

- fact_MatchResult
- fact_TeamMatchStats

- fact_LeagueSnapshot

Dimension Tables:

- dim_Time
- dim_Team
- dim_Season
- dim_Referee
- dim_Location



5. ETL Process

A robust **Extract, Transform, Load (ETL)** process forms the backbone of the Premier League Analytics System, ensuring clean, structured, and analysis-ready data flows from the source CSVs into both OLTP and OLAP databases.

The ETL was implemented using **Python**, leveraging libraries such as pandas, NumPy, and MySQL-connector-python for efficient data processing and SQL interaction.

5.1 Extraction

- Reads 10 seasonal CSV files sourced from football-data.co.uk
- Parses key fields: match metadata, team names, referees, scores, and odds
- Logs file metadata to ensure completeness and traceability

5.2 Transformation

- **Cleaning**: Handles missing values, invalid types, and standardizes categorical fields (e.g., referee names, team IDs)
- **Derivation**: Computes additional fields like goal difference, total cards, and aggregated odds
- **Mapping**: Resolves foreign key relationships to dimension tables (e.g., teams, referees, seasons)

5.3 Loading

- Loads the cleaned and validated data into the **OLTP database**, populating:
  - Matches, MatchStatistics, Bookmakers, BettingOdds
  - Lookup tables like Teams, Seasons, Referees, etc.
- A stored procedure (refresh_dwh_prc) is used to populate the **OLAP database** from the OLTP, executing steps like:
  - Truncating and reloading fact/dimension tables
  - Joining operational tables to derive analytical facts
  - Maintaining surrogate keys and ensuring dimension conformance

5.4 Automation & Reusability

- ETL notebooks and scripts are modular, supporting batch reprocessing
- The pipeline is resilient to schema drift due to pre-validation of headers and data types

5.5 Implementation Workflow & User Journey

The system follows a specific implementation sequence that combines automated setup with user-driven operations:

Phase 1: Initial Setup (Jupyter Notebooks)

1. Execute `01_Sch_Squ_OPT_DB_Setup.ipynb` to create the operational database schema
2. Run analytical database setup notebooks (`02_setup_analytical_db.ipynb` through `05_etl_update_job.ipynb`)

Phase 2: Operational Usage (GUI Application)

1. Launch `main.py` to start the PyQt5 GUI application
2. Login with credentials (Username: Admin, Password: admin)
3. Navigate to **Views → ETL Control** to access the data upload interface
4. Upload CSV files through the staging interface, which automatically triggers data validation and loading

Phase 3: Analytics & Reporting (Hybrid Approach)

- **GUI-based Analytics**: Interactive dashboards for team trends, referee analysis, and betting odds evaluation
- **Notebook-based Analytics**: Advanced statistical analysis and custom visualizations

Critical Integration Point: The GUI's ETL Control module serves as the **primary interface** for ongoing data operations, while notebooks provide the foundation and advanced analytical capabilities.

6. Operational Module

The Operational Module of the Premier League Analytics System manages structured transactional data related to football matches, including team performance, referee assignments, and betting odds. It ensures data accuracy through strict normalization and supports real-time insertion and updates.

6.1 Core Functionalities

The OLTP schema supports the following key operations:

- **Match Management**: Insertion of match metadata, including home/away teams, scores, dates, and referees

- **Team & Referee Records**: Auto-mapped foreign keys for teams and referees ensure consistency across seasons

- **Season & Division Mapping**: Each match is linked to a specific season and division, enabling year-wise analysis

- **Match Statistics**: Detailed metrics like goals, shots, fouls, possession, and cards are stored in MatchStatistics
- **Betting Odds Integration**: Odds from multiple bookmakers are stored per match via the Bookmakers, Markets, and BettingOdds tables, allowing comparative analytics

Referential integrity is maintained via primary/foreign key constraints, and transactional accuracy is ensured using batch commits during data loads.

6.2 Graphical User Interface (GUI) - Primary Operational Interface

The PyQt5-based GUI serves as the **central hub** for system operations, bridging the gap between technical database setup and user-friendly data management.

Authentication & Access Control:

- Secure login system (Default credentials: Admin/admin)
- Role-based access to different system components

Core GUI Modules:

- **ETL Control Panel** *(Primary Feature)*:
  - CSV file upload and validation
  - Staging table management
  - Real-time ETL job monitoring
  - Data quality reporting
- **Analytics Dashboard**: Interactive visualizations accessible post-ETL
- **Database Management**: Connection status and operational controls

Integration with Notebooks: While Jupyter notebooks handle schema creation and advanced analytics, the GUI provides the operational bridge for daily data management tasks.

7. Analytical Module

The Analytical Module is designed to extract strategic insights from historical football match data by leveraging a **star schema**-based OLAP database. This module supports advanced querying across multiple dimensions such as season, team, location, and referee, and is optimized for trend and performance analysis.

7.1 Analytical Schema Design

The OLAP database includes several **fact** and **dimension** tables, enabling flexible slicing and dicing of match data:

Fact Tables:

- fact_MatchResult: Contains aggregated match-level outcome data (home/away goals, results)

- fact_TeamMatchStats: Includes granular team-level performance metrics (shots, fouls, possession, etc.)
- fact_LeagueSnapshot: Periodic snapshots of league standings or KPIs by round/week

Dimension Tables:

- dim_Team: Describes each club with attributes like name, code, and division
- dim_Referee: Stores referee information and regional mapping
- dim_Season: Captures the year and associated metadata
- dim_Location: Represents the home stadium or venue for matches
- dim_Time: Supports temporal queries by day, week, month, quarter, and year

These tables are linked by surrogate keys and maintained through a stored procedure that refreshes the OLAP database using data from the operational system.

7.2 Analytical Use Cases

Managers and analysts can use the OLAP module to answer questions such as:

- How do home and away performances differ by team and season?
- Which referees are associated with the most fouls or cards?
- Which teams consistently outperform others in possession or shot accuracy?
- What are the seasonal trends in match outcomes across the league?
- Queries are executed via **Jupyter Notebooks**, allowing users to:
- Run predefined or custom SQL statements
- Visualize data using Python plots
- Export reports for further analysis or presentation

8. Analytical Insights

The analytical capabilities of the Premier League Analytics System are showcased through an interactive dashboard that enables exploration of team performance trends, betting odds accuracy, and bookmaker behaviour. These insights are visualized using data pulled from the OLAP schema and transformed into interpretable charts using Python and PyQt5 GUI components.

8.1 Team Trend Analysis

This use case focuses on evaluating how teams accumulate points over the course of a season, match results by round, and general consistency in performance.

Arsenal's – Cumulative Points Tracker (2022–23): The chart below shows the cumulative points earned by Arsenal during the 2022–23 season.

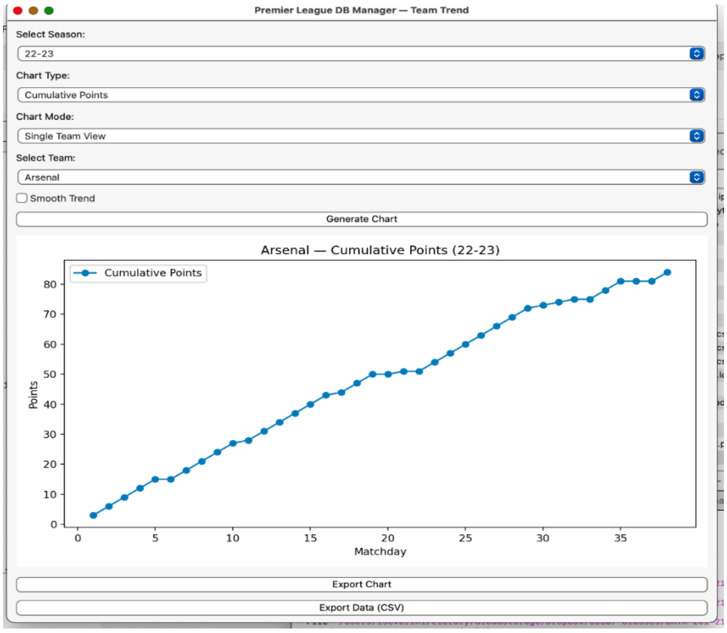


Figure 8.1: Arsenal's match-by-match cumulative point growth

Team – Match Results (W/D/L) Overview

This chart visualizes each match result as a Win (W), Draw (D), or Loss (L) per round, providing clarity on consistency and streaks.

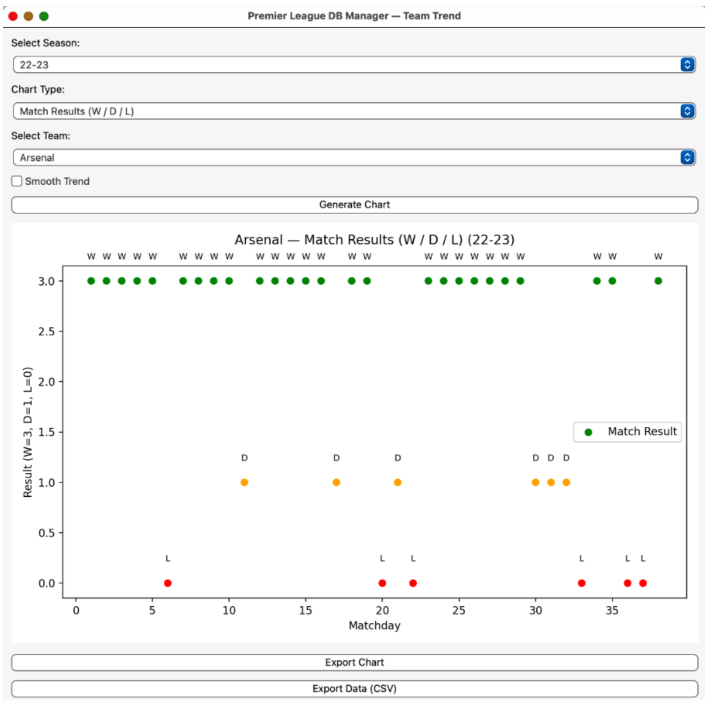


Figure 8.2: Arsenal's 2022–23 match result trend

8.2 Odds Analysis – Bookmaker vs Reality

This set of visualizations focuses on evaluating the accuracy and margin behaviour of the bookmakers, using both implied and actual outcome data.

Implied vs Actual – Match Result Probabilities

This chart compares the probabilities implied by betting odds with the actual outcome rates across all matches.

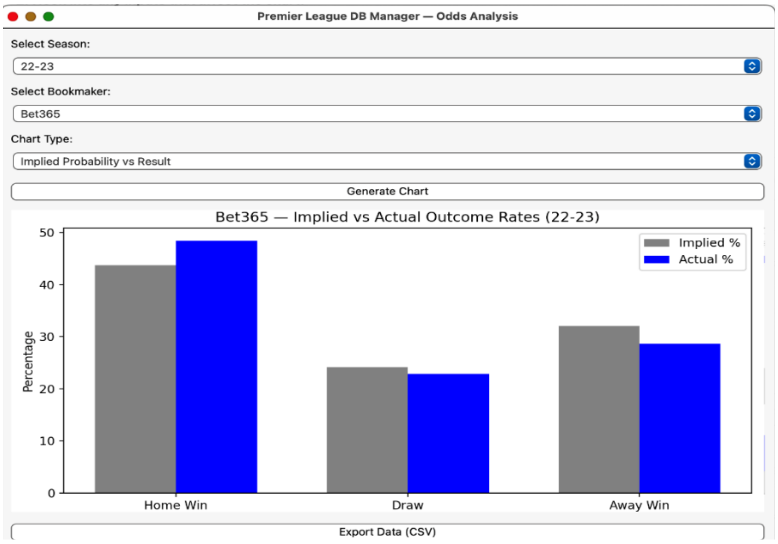


Figure 8.3: Home win, draw, and away win probabilities

Over/Under Goal Market – Implied vs Actual

This visual shows how often the bookmaker's implied odds for "Over 2.5 goals" aligned with reality.

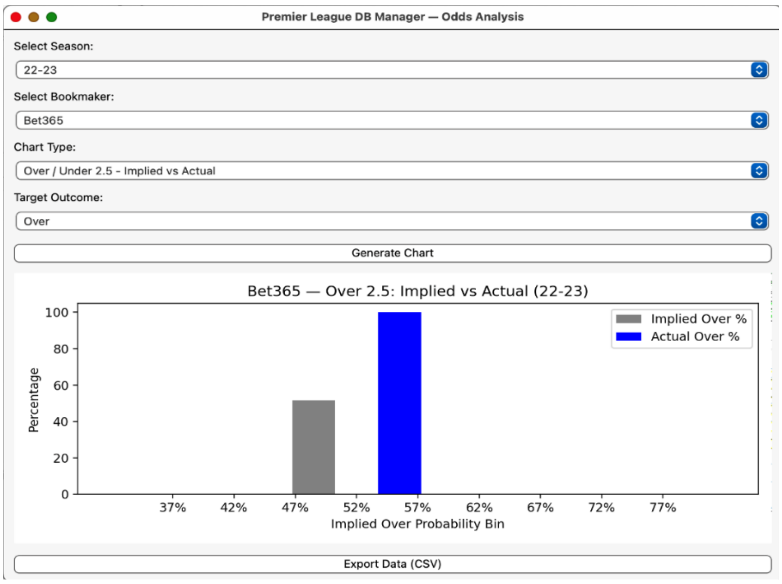


Figure 8.4: Over 2.5 Goal Line – Implied vs Actual Outcome

Bookmaker Margin Per Match

This graph illustrates the margin retained by the bookmaker on each match based on odds set, offering a lens into profitability and risk management.

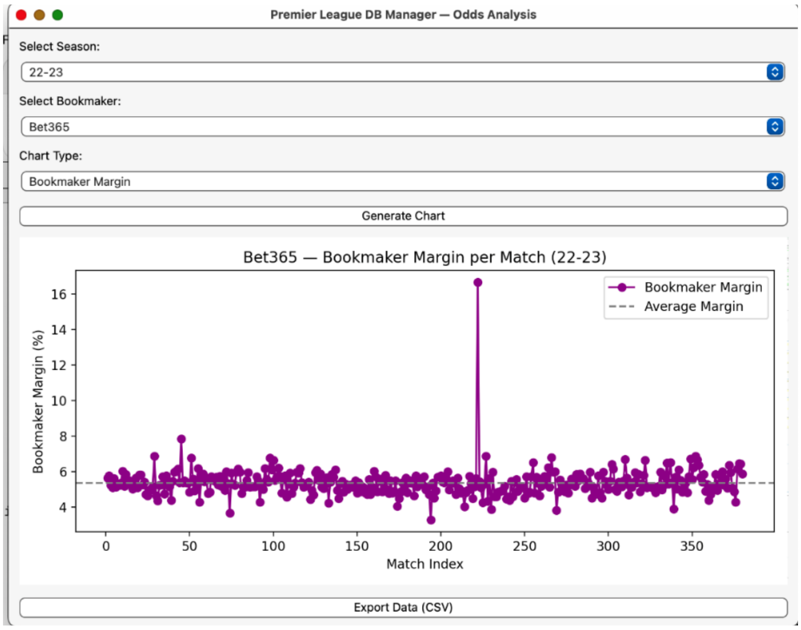


Figure 8.5: Bet365's margin spread across all fixtures

8.3 Seasonal Performance Breakdowns

- Aggregated KPIs like:
    - Points per team
    - Goal difference per Team
    - Win/loss streaks are tracked across seasons to measure league competitiveness and team consistency.
- Monthly/quarterly rollups support temporal trend analysis and anomaly detection.

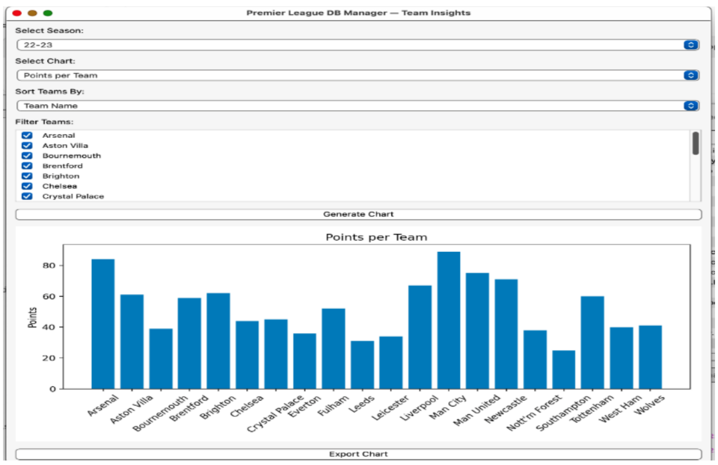


Figure 8.6: Referee match activity trend

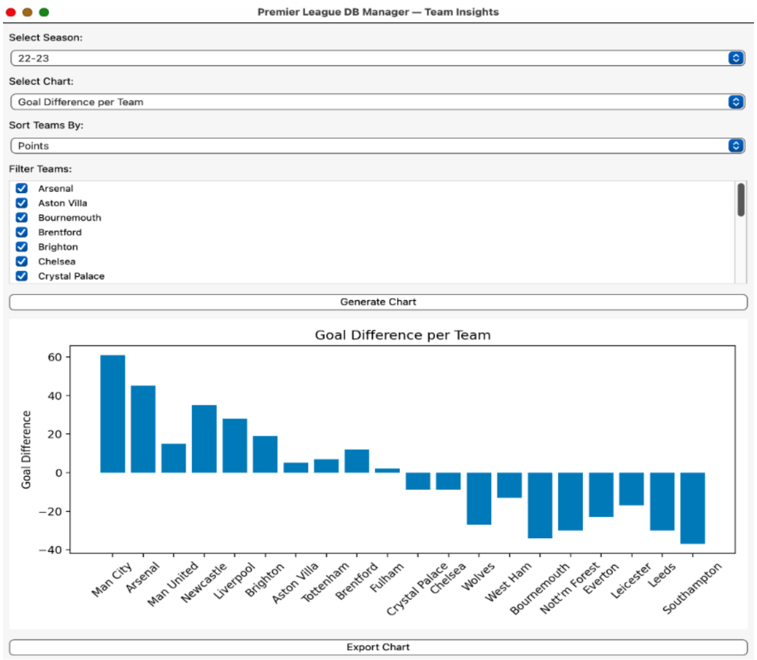


Figure 8.7: Goal Difference per team in season 22 – 23

Figure 8.7: Win/Loss streaks per team in season 22 - 23

8.4 Referee Influence Analysis

Referees significantly impact the flow and discipline of football matches through their decisions on fouls and cards. This analysis focuses on tracking referee behavior using metrics such as yellow cards, red cards, and fouls across matchdays and seasons.

Data from the operational and analytical databases is used to generate trends and comparisons that highlight variations in officiating styles.By visualizing individual referee patterns and comparing them across the league, this module helps identify tendencies such as strictness.These insights enable teams to better anticipate match conditions and offer a data-driven perspective on referee consistency and performance throughout the season.

Referee Trends Over Time – D Bond (2022–23)

The following visualization illustrates the number of fouls, yellow cards, and red cards issued by referee D Bond over a rolling set of matchdays in the 2022–23 season.

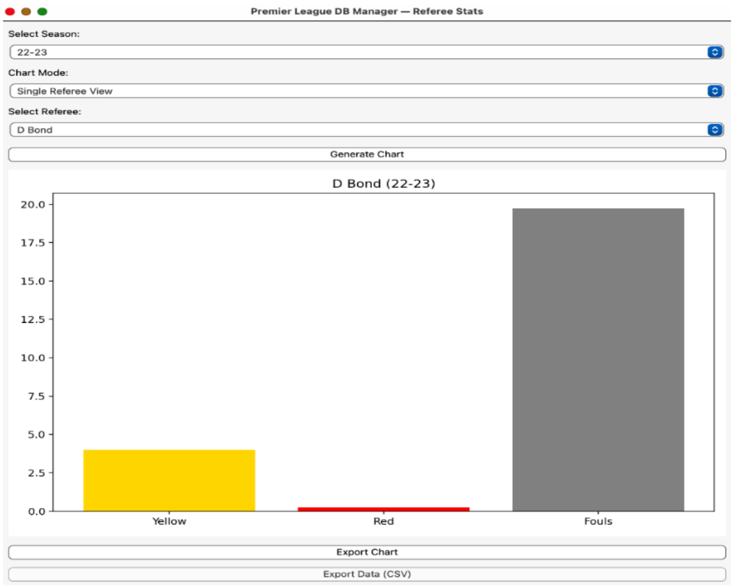


Figure 8.8: Referee match activity trend

Yellow Cards Per Referee

This chart presents an overview of yellow card issuance rates per match, helping identify stricter referees across the league.

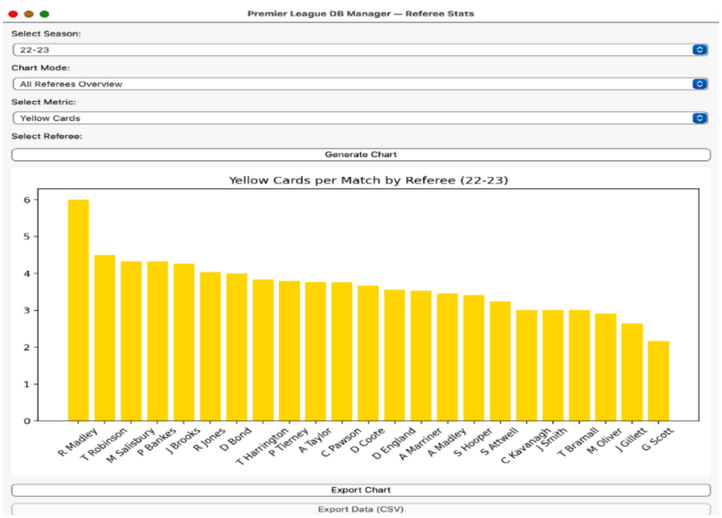


Figure 8.9: League-wide yellow cards per referee

Referee Comparison – D Bond vs A Madley

A side-by-side comparison of two referees over the season reveals differences in disciplinary style and foul frequency.

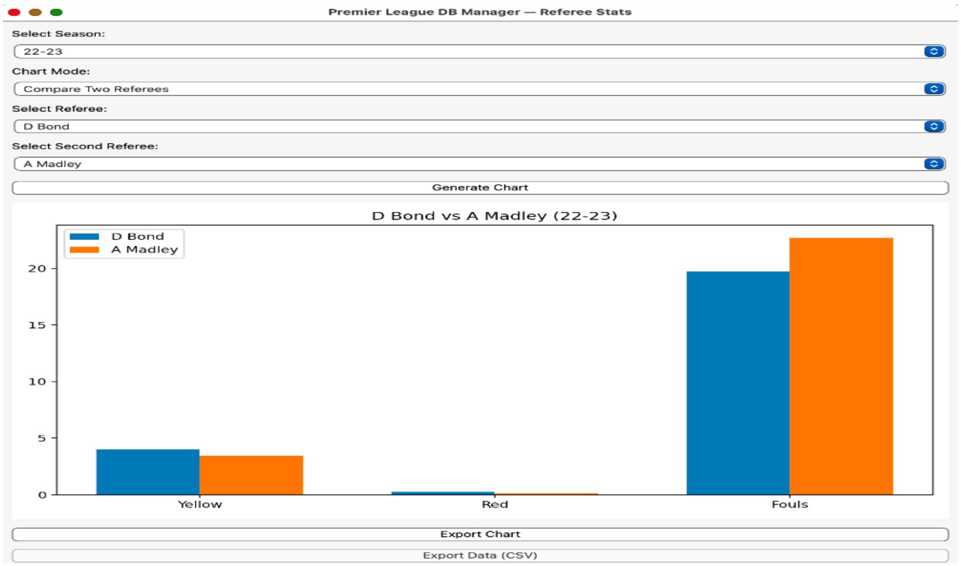


Figure 8.8: Carding and foul trend: D Bond vs A Madley

These insights are derived from **pre-aggregated fact tables** in the star schema and visualized using Python tools like matplotlib and seaborn. Analysts can drill down or roll up based on season, team, referee, or location to support hypothesis testing or data storytelling.

9. Technical Challenges & Solutions

During the development of the Premier League Analytics System, several technical challenges were encountered while handling raw data, designing normalized and star schemas, and ensuring ETL robustness. The following solutions were implemented to resolve these issues efficiently.

9.1 Reserved Keywords in SQL

- **Challenge**: Some CSV column names conflicted with MySQL reserved keywords (e.g., match, time, rank)
- **Solution**: All problematic identifiers were wrapped in backticks (`) in SQL queries to avoid syntax errors

9.2 Inconsistent CSV Schema Across Seasons

- **Challenge**: Column order, naming, and available metrics varied slightly across CSVs
- **Solution**: A flexible column-mapping logic was implemented in the ETL script to:
    - Normalize column names
    - Add default values for missing metrics
    - Skip irrelevant or inconsistent fields safely

9.3 Referential Integrity & Foreign Key Conflicts

- **Challenge**: Loading data into the operational database caused integrity errors when referenced dimension data (e.g., teams, referees) was missing
- **Solution**: Dimension tables (Teams, Referees, etc.) were preloaded and validated before inserting transactional records. ETL scripts included fallback ID lookups and logging of unmatched references

9.4 OLAP Star Schema Loading Performance

- **Challenge**: Populating OLAP fact tables from OLTP caused slowdowns during large batch loads
- **Solution**:
    - Created a **stored procedure** (refresh_dwh_prc) to truncate and reload OLAP tables atomically
    - Indexed surrogate keys and fact relationships
    - Used pre-joins and aggregated selects to minimize transformation steps during load

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9.5 Betting Data Integration

- **Challenge**: Bookmaker odds data varied in completeness and format across matches
- **Solution**:
    - Created normalized structures for Bookmakers, Markets, and BettingOdds
    - Used conditional inserts to only store valid odds
    - Implemented fallback logic when markets were missing

These strategies ensured that the system remains robust, extensible, and efficient even when working with complex historical datasets and dynamic analytical requirements.

10. **Technical Aspects**

The following is the list of the files used in the application:

Implementation Note: Hybrid Architecture

```
🔧 CRITICAL IMPLEMENTATION SEQUENCE:
1. Notebooks: Database schema setup (OLTP & OLAP)
2. GUI (main.py): Operational data management & ETL
3. Notebooks + GUI: Analytics and reporting

The GUI's ETL Control module is the PRIMARY interface for ongoing
data operations, making the system accessible to non-technical users
while maintaining the analytical power of Jupyter notebooks.
```

Operational Module

S.No	File Name	Type	Description
1	Sch_Squ_OPT_DB_Setup.ipynb	Jupyter NB	Creates the schema for the operational (OLTP) database
2	Folder: pl_gui	Python	Loads raw data into the operational database/ Anlytics Options
3	data201.py	Python	Reusable functions for DB connection, transformation, etc.
4	premier_league_analytics.ini	Configuration	Database connection configuration file

Analytical Module

S.No	File Name	Type	Description
5	2_setup_analytical_db.ipynb	Jupyter NB	Defines and creates the analytical (OLAP) schema
6	3_load_analytical_data.ipynb	Jupyter NB	Loads dimension and fact tables using data from OLTP
7	4_premier_league_analytics.ipynb	Jupyter NB	Executes analytical queries and visualizations
8	5_etl_update_job.ipynb	Jupyter NB	Refreshes OLAP data using the stored procedure

Documentation & Support

S. No	File Name	Type	Description
09	Proposal.txt	Text	Initial project proposal
10	Premier_League_Analytics_Project_Presentation.pptx	PowerPoint	Final presentation
11	Premier_League_Analytics_Project_Presentation.pdf	PDF	Presentation PDF version
12	RemoteDatabaseServer.pdf	PDF	Info about database server hosting
13	README.md	Markdown	Repository instructions and overview

12. Database Technical Details

12.1. DB Objects

Database Type	Object Type	Name	Usage
Operational	Table	Teams	Metadata for clubs
Operational	Table	Seasons	Premier League seasons
Operational	Table	Referees	Match referee data
Operational	Table	Matches	Core match records
Operational	Table	MatchStatistics	Possession, shots, cards, etc.
Operational	Table	Bookmakers	List of betting operators
Operational	Table	Markets	Betting market types (Home, Draw, Away)
Operational	Table	BettingOdds	Odds values for each market and match

Database Type	Object Type	Name	Usage
Analytical	Table	dim_Team	Dimension table for team details
Analytical	Table	dim_Season	Year-wise dimension
Analytical	Table	dim_Referee	Referee dimension table
Analytical	Table	dim_Location	Stadium or match venue info
Analytical	Table	dim_Time	Time hierarchy dimension
Analytical	Fact Table	fact_MatchResult	Aggregated result facts
Analytical	Fact Table	fact_TeamMatchStats	Performance facts per team
Analytical	Fact Table	fact_LeagueSnapshot	League position and points by round
Analytical	Stored Proc	refresh_dwh_prc	Refreshes OLAP DB with data from OLTP

12.2. ETL

The ETL process is driven by Jupyter notebooks and Python scripts to ingest and transform seasonal match data.

- **Operational**:
  Files from football-data.co.uk are loaded into the OLTP schema. Data is normalized, and foreign keys (teams, referees, seasons) are validated during insertion.
- **Analytical**:
  A stored procedure named refresh_dwh_prc truncates OLAP fact/dimension tables and repopulates them using aggregated joins across the OLTP tables. Time dimension is derived from match dates.

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13. Queries (Operational DB)

Purpose	Query
Match Info	SELECT * FROM Matches WHERE match_id = 1024;
List Teams	SELECT team_id, team_name FROM Teams;

Purpose	Query
Odds for a Match	SELECT b.name, m.type, o.home_odds, o.draw_odds, o.away_odds FROM BettingOdds o JOIN Bookmakers b ON o.bookmaker_id = b.bookmaker_id JOIN Markets m ON o.market_id = m.market_id WHERE o.match_id = 1024;
Match Statistics	SELECT * FROM MatchStatistics WHERE match_id = 1024;

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14. Queries (Analytical DB)

Purpose	Query
Top 10 Goal-Scoring Teams	SELECT t.name, SUM(m.home_goals + m.away_goals) AS total_goals FROM fact_MatchResult m JOIN dim_Team t ON m.team_id = t.team_id GROUP BY t.name ORDER BY total_goals DESC LIMIT 10;
Average Possession by Season	SELECT s.year, AVG(ts.possession_home) AS avg_possession FROM fact_TeamMatchStats ts JOIN dim_Season s ON ts.season_id = s.season_id GROUP BY s.year;
Referee Card Average	SELECT r.name, AVG(ts.cards) AS avg_cards FROM fact_TeamMatchStats ts JOIN dim_Referee r ON ts.referee_id = r.referee_id GROUP BY r.name ORDER BY avg_cards DESC;

15. Conclusion

The Premier League Analytics System is a comprehensive solution integrating normalized transactional processing with analytical reporting capabilities tailored for football data. Through effective use of Python, MySQL, PyQt5, and Jupyter, the project successfully achieved the following:

- Built a dual-database model with OLTP and OLAP components
- Extracted and normalized over a decade of Premier League data
- Integrated betting odds alongside match performance data
- Enabled advanced visual and SQL-based analytics using a star schema
- Delivered a modular and extensible framework with GUI and ETL

Hybrid Implementation Success: A key achievement of this project is the successful integration of Jupyter notebooks for technical setup with a user-friendly GUI for operational management. This hybrid approach ensures that technical users can leverage the full power of Python and SQL for complex analytics, while operational users have an intuitive interface for daily data management tasks. The GUI's ETL Control module particularly demonstrates how complex data engineering processes can be made accessible through thoughtful interface design.

This project highlights strong competencies in database design, data engineering, and sports analytics — translating classroom concepts into a real-world solution that can support club decision-makers, analysts, and fans alike.